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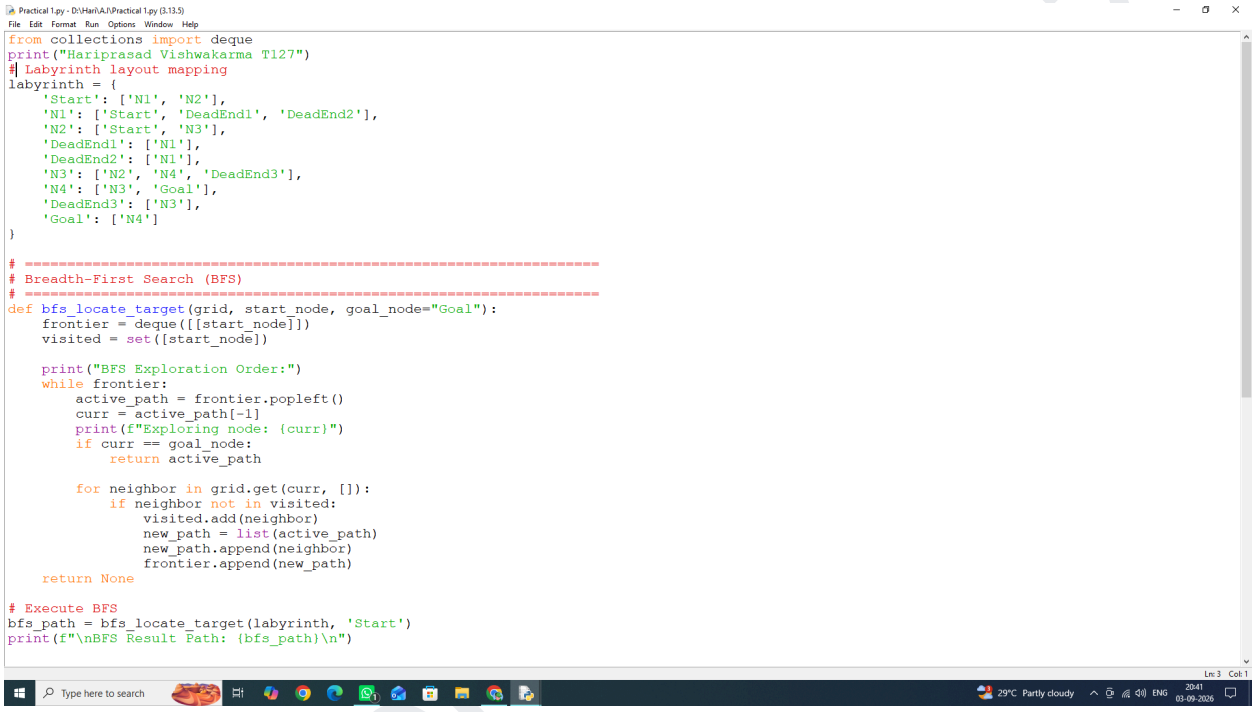
A.I. Practical No.9

Aim :- A) Implement the Breadth First Search algorithm to solve a given problem.

B) Implement the Iterative Depth First Search algorithm to solve the same problem.

C) Compare the performance and efficiency of both algorithms.

Code:-



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```
                visited.add(neighbor)
                new_path = list(active_path)
                new_path.append(neighbor)
                frontier.append(new_path)
    return None

# Execute BFS
bfs_path = bfs_locate_target(labyrinth, 'Start')
print(f"\nBFS Result Path: {bfs_path}\n")


# ======================================================================
# Iterative Deepening Depth-First Search (IDDFS)
# ======================================================================
def depth_limited_search(grid, curr, goal_node, depth_rem, active_path):
    if curr == goal_node:
        return active_path
    if depth_rem <= 0:
        return None

    for neighbor in grid.get(curr, []):
        if neighbor not in active_path:
            new_path = list(active_path)
            new_path.append(neighbor)
            res = depth_limited_search(grid, neighbor, goal_node, depth_rem - 1, new_path)
            if res is not None:
                return res
    return None

def iddfs_locate_target(grid, start_node, goal_node="Goal", max_depth=10):
    print("IDDFS Iterations:")
    for limit in range(max_depth):
        print(f"Searching with depth limit: {limit}")
        res = depth_limited_search(grid, start_node, goal_node, limit, [start_node])
        if res is not None:
            return res
    return "No path found within max depth limit"

# Execute IDDFS
iddfs_path = iddfs_locate_target(labyrinth, 'Start')
print(f"\nIDDFS Result Path: {iddfs_path}")
```

Output:-

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: D:\Hari\A.I\Practical 1.py ==========
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BFS Exploration Order:
Exploring node: Start
Exploring node: N1
Exploring node: N2
Exploring node: DeadEnd1
Exploring node: DeadEnd2
Exploring node: N3
Exploring node: N4
Exploring node: DeadEnd3
Exploring node: Goal

BFS Result Path: ['Start', 'N2', 'N3', 'N4', 'Goal']

IDDFS Iterations:
Searching with depth limit: 0
Searching with depth limit: 1
Searching with depth limit: 2
Searching with depth limit: 3
Searching with depth limit: 4

IDDFS Result Path: ['Start', 'N2', 'N3', 'N4', 'Goal']
>>>
```

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